C.5. An interesting detour: Grokking for Fixed-size AI Dataset.

Now consider the scenario where the AI synthesized dataset has fixed size T_{AI} (e.g a frozen chunk of the web), while the clean dataset size is a scalable parameter T_{real}. Taking $T = T_{real} + T_{AI}$ and $\pi = T_{real}/T$, we have the following corollary of Theorem 3.2, which includes Corrolary 3.3.

Corollary C.4. *We have the following.*

(A) Early-Stage Dynamics. *For $T_{real} \ll k^{\beta}$, it holds that*

$$E_{test} \asymp (T_{real} + T_{AI})^{-(1-1/\beta)} + k^{-(\beta-1)} \tag{32}$$

(B) Later-Stage Dynamics. *As soon as $T_{real} \geq Ck^{\beta}$ (where C is an absolute constant), it holds that*

$$E_{test} \asymp T_{real}^{-(1-1/\beta)}. \tag{33}$$

As mentioned in Section 3, AI synthesized data is helpful in the regime where real data is scarce. Once more of real data becomes available the model grokks for a while and then forgets the AI synthesized data to recover the normal scaling law w.r.t T_{real}. Figure 20 gives an illustration of this phenomenon in various settings.

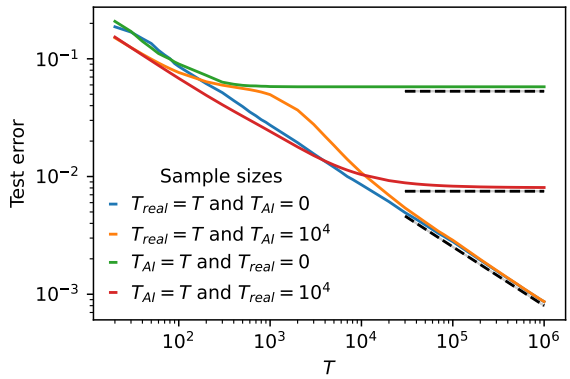

Figure 20. **Hutter LLM.** true distribution of the data is Zipf with exponent $\beta = 2$. Here, the scalable resource is either clean data or AI data-generated data, corresponding to a version of real data with its tail cut at rank k (here we use $k = 10$). We either mix with a fixed amount (here $T' = 10^4$ samples) of the other resource, or we don't mix at all. Then we scale up the scalable resource by cranking up T. As predicted by Corollary 3.3, the orange curve always grokks: AI synthesized data is helpful in the regime where real data is scarce; once more of real data becomes available the model grokks for a while and then forgets the AI synthesized data. Note that the green curve (only AI data) and red curve (AI + real data) don't grokk because the optional resource (real data) is not being scaled; if it is also scaled, then green and red will provably grokk (as in Figure 3). The diagonal broken line corresponds to the standard Hutter scaling law $E_{test} \asymp T^{-c}$, where $c := 1 - 1/\beta$. The horizontal broken lines correspond to $E_{test} \asymp k^{-(\beta-1)}$ and $E_{test} \asymp T'^{-c}$, both predicted by Theorem 2.1.

C.6. Proof of Theorem 3.1

Note that explicitly,

$$\pi_i \asymp \begin{cases} N^{\alpha} p_i, & \text{if } i \geq N, \\ 0, & \text{else,} \end{cases} \tag{34}$$

where $\alpha := \beta - 1$. This is because the normalization constant is $\sum_{i \geq N} p_i = \sum_{i \geq N} i^{-\beta} \asymp N^{-\alpha}$. Now, mix this distribution with q with equal weights $1/2$, to obtain a new distribution

$$\begin{aligned} q'_i = q_i/2 + \pi_i/2 &= \begin{cases} q_i/2, & \text{if } i \leq k, \\ \pi_i/2, & \text{if } k \geq N, \\ 0, & \text{otherwise} \end{cases} \\ &\asymp \begin{cases} p_i, & \text{if } i \leq k, \\ N^{\alpha} p_i, & \text{if } k \geq N, \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \tag{35}$$

For simplicity, assume $N \geq k+1$ (otherwise, we have all of p). Build a "Hutter" LLM from an iid sample of size T from this distribution (this is equivalent to mixing T samples from q and T samples from π. Then, it is easy to see that the test error is given by

$$E_{test} = \sum_{i \geq 1} p_i (1-q_i')^T \asymp \sum_{1 \leq i \leq k} p_i (1-p_i)^T + \sum_{k+1 \leq i \leq N-1} p_i + \sum_{i \geq N} p_i (1 - N^\alpha p_i)^T. \tag{36}$$

Thanks to previous computations, we know that for large k, N, and T

- The first sum is of order $T^{-c}\left(\Gamma(c, Tk^{-\beta}) - \Gamma(c,T)\right) = O(T^{-c})$.
- The third sum is of order $T^{-c}\left(\Gamma(c,0) - \Gamma(c, TN^\alpha N^{-\beta})\right) = T^{-c}\left(\Gamma(c,0) - \Gamma(c,TN)\right) \asymp T^{-c}$.
- The second sum is of order $k^{-\alpha} - N^{-\alpha} = ((\frac{N}{k})^\alpha - 1)N^{-\alpha}$, where $\alpha := \beta - 1$.

We deduce that

$$E_{test} \asymp T^{-c} + \left(\left(\frac{N}{k}\right)^\alpha - 1\right) N^{-\alpha}, \text{ for large } k, N, T, \tag{37}$$

and the result follows. □

D. Proofs for the Tailed Bigram Model (Section 4)

D.1. Warm-up: Revisiting the Classical Hutter Setup

As a sanity check, with the framework of Equation (17), let us momentarily consider the non-autoregressive setup where $p(\cdot \mid i) = \delta_{y_i}$ for all i, as in classical Hutter. Then, an easy computation shows that

$$TV(q_T(\cdot \mid i), p(\cdot \mid i)) = 1 - q_T(y_i \mid i) + \sum_{j \neq y_i} q_T(j \mid i) = 2(1 - q_T(y_i \mid i)).$$

Now, by construction, $q_T(y_i \mid i) = 1[i \in \mathcal{D}_T]$. Thus,

$$\mathbb{E}\left[1 - q_T(y_i \mid i)\right] = \mathbb{P}(i \notin \mathcal{D}_T) = (1-p_i)^T.$$

We deduce that

$$\mathbb{E}\left[TV(q_T(\cdot \mid i), p(\cdot \mid i))\right] = 2(1-p_i)^T.$$

Therefore,

$$\begin{aligned} E_{test} &= \sum_i p_i \mathbb{E}\left[TV(q_T(\cdot \mid i), p(\cdot \mid i))\right] \\ &= 2\sum_i p_i (1-p_i)^T \asymp T^{-(1-1/\beta)}, \end{aligned} \tag{38}$$

and we recover the classical Hutter result! Thus, our test metric defined in (17) is pointing in the right direction, conceptually.

D.2. Proof of Theorem 4.1

The proof will be based on the results of (Berend & Kontorovich, 2012). ()

Upper-Bound. Observe that for any choice of mappings $\pi_1, \pi_2, \ldots$, we have

$$a_T(i) := \sum_{j \mid p(j|i) \leq 1/n_T(i)} p(j \mid i) \asymp \sum_{j \mid \pi_i(j) \geq n_T(i)^{1/\beta}} \pi_i(j)^{-\beta} \leq \sum_{k \mid k \geq n_T(i)^{1/\beta}} k^{-\beta} \asymp n_T(i)^{-(1-1/\beta)}$$
$$b_T(i) := n_T(i)^{-1/2} \sum_{j \mid p(j|i) \geq 1/n_T(i)} \sqrt{p(j \mid i)} \asymp n_T(i)^{-1/2} \sum_{j \mid \pi_i(j) \leq n_T(i)^{1/\beta}} \pi_i(j)^{-\beta/2}$$
$$\lesssim n_T(i)^{-1/2} \sum_{k \mid k \leq n_T(i)^{1/\beta}} k^{-\beta/2} \asymp n_T(i)^{-c}.$$

We deduce that $c_T(i) := a_T(i) + b_T(i) \lesssim n_T(i)^{-c}$ for any i. Importantly, the hidden constants don't depend on i. Therefore, thanks to [Lemma 9] (Berend & Kontorovich, 2012), we have

$$E_{test} \leq \sum_i p_i \mathbb{E}\left[c_T(i)\right] \lesssim \sum_i p_i \mathbb{E}\left[n_T(i)^{-c}\right] \overset{(*)}{\leq} \sum_i p_i (\mathbb{E}\left[n_T(i)\right])^{-c} = \sum_i p_i (T p_i)^{-c} = T^{-c} \sum_i p_i^{1-c} \lesssim T^{-c}, \tag{39}$$

where we have used Jensen's inequality in (*), since the function $x \mapsto x^{-c}$ is concave.

Lower-Bound. WLOG[6] consider the following specific choice of permutations defined by $\pi_i(j) = j$ (i.e doesn't depend on i). Then,

$$a_T(i) = \sum_{j \geq n_T(i)^{1/\beta}} j^{-\beta} \asymp n_T(i)^{-(1-1/\beta)},$$
$$b_T(i) = n_T(i)^{-1/2} \sum_{j \leq n_T(i)^{1/\beta}} j^{-\beta} \asymp n_T(i)^{-c}.$$

Thanks to the definition of E_{test} and [Proposition 5] (Berend & Kontorovich, 2012), we deduce that if $\beta \in (1, 2)$, then

$$E_{test} \geq \sum_i p_i \mathbb{E}\left[(a_T(i) + b_T(i) - n_T(i)^{-1/2})\right] \asymp \sum_i p_i \mathbb{E}\left[n_T(i)^{-c} - n_T(i)^{-1/2})\right] \asymp \sum_i p_i \mathbb{E}\left[n_T(i)^{-c}\right], \tag{40}$$

i.e $E_{test} \gtrsim \sum_i p_i \mathbb{E}\left[n_T(i)^{-c}\right]$. Now, since $n_T(i) \sim Bin(T, p_i)$, standard Binomial concentration arguments tell us that $n_T(i) \leq 1.5 T p_i$ w.p $1 - e^{-C p_i T}$, where C is an absolute constant. We deduce that

$$E_{test} \gtrsim \sum_i p_i (1.5 T p_i)^{-c} (1 - e^{-C p_i T}) \asymp T^{-c} \sum_i p_i^{1-c} - T^{-c} \underbrace{\sum_i p_i^{1-c} e^{-C p_i T}}_{o(1)} \asymp T^{-c},$$

which completes the proof. □

D.3. Proof of Theorem 4.2

It suffices to replace $n_T(i)$ in (39) and (40) of the proof of Theorem 4.1 with $n_T(i) \wedge k^\beta$, and use the elementary fact that $(n_T(i) \wedge k^\beta)^{-c} = n_T(i)^{-c} \vee k^{-\beta c} \asymp n_T(i)^{-c} + k^{-\beta c}$. The rest of the proof proceeds as that of Theorem 4.1.

D.4. Extensions

Note that the above setup can be extended to the following

$$p(j \mid i) = \rho(\pi_i(j)),$$

where ρ is a distribution on $\mathbb{N}_*$. In particular, taking $\rho(z) \propto z^{-\beta}$, recovers the setup considered above. It is clear that mechanics of the proof of Theorem 4.1 should be applicable here, leading to scaling laws which depend explicitly on ρ.

[6] A summable series of nonnegative numbers (like in $a_T(i)$ and $b_T(i)$) can be reordered without changing the value.